Microeconomics

The consumer problem

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From preferences to an actual choice

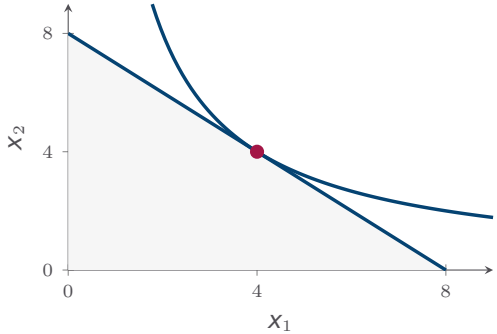
Last time: what people want, what they can afford, and how the two fit together.

Today: existence and uniqueness, Marshallian demand, the Lagrangian and Kuhn–Tucker conditions.

Then: indirect utility, the envelope theorem and Roy's identity.

A separate reserve section introduces expenditure minimisation and Hicksian demand for the next course block.

The picture we already know



$u(x) = \sqrt{x_1 x_2}$, $p = (1, 1)$, $m = 8$. Which affordable bundle is best? $(4, 4)$. Today we want a systematic method, not just a convincing picture.

The consumer problem

Let $X = \mathbb{R}^L_+$, prices $p \in \mathbb{R}^L_{++}$ and wealth $m > 0$.

$$B(p, m) = \{x \geq 0 : p \cdot x \leq m\}.$$

$$\max_{x \in B(p,m)} u(x).$$

Preferences enter through u; prices and wealth enter through the feasible set.

What did we assume when we wrote that?

Price-taking: prices are given rather than negotiated.

Linear prices: every unit of a good has the same unit price.

Divisibility: quantities can vary continuously.

A specified decision horizon: if saving is possible, future consumption or saving must enter the model.

What changes for a subscription, a quantity discount or an indivisible purchase?

Two useful facts about the budget

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Currency invariance: for every $t > 0$, $B(tp, tm) = B(p, m)$.
Nonemptiness: the zero bundle is affordable.
Compactness: the budget is closed, and $0 \leq x_i \leq m/p_i$ for every i.
Strictly positive prices give a finite upper bound on every quantity.

Does an optimum exist?

Assumptions: u is continuous on $\mathbb{R}^L_+$, $p \gg 0$ and $m > 0$.

The budget is nonempty and compact.

By the extreme value theorem, a continuous function attains a maximum on it.

Existence does not yet imply uniqueness, an interior solution or differentiability of demand.

Which assumption is doing the work?

Suppose $p_1 = 0$, $p_2 = 1$, $m = 1$, and $u(x) = x_1$.

Every quantity of good 1 is affordable. Utility is unbounded above, so there is no maximiser.

Zero prices do not always destroy existence, but compactness is no longer guaranteed.

Can you give a case with multiple maximisers even though the budget is compact?

Marshallian demand is initially a correspondence

$$D(p, m) = \underset{x \geq 0,\, p \cdot x \leq m}{\arg\max}\, u(x).$$

$D(p, m)$ contains **all** optimal bundles. It may contain more than one.
When it is a singleton, write its unique element as $x(p, m)$: a demand function.
It is the earlier choice rule applied specifically to budget menus:
$D(p, m) = C(B(p, m))$.

Demand is homogeneous of degree zero

For every $t > 0$, $D(tp, tm) = D(p, m)$.
The feasible set and utility ranking are unchanged, so the set of optimisers is unchanged.

$$p = (2, 1),\ m = 12 \quad \text{versus} \quad p = (200, 100),\ m = 1200.$$

This is a change of currency units, not an increase in purchasing power.

Budget exhaustion: Walras' law

If preferences are locally nonsatiated, every $x^* \in D(p, m)$ satisfies $p \cdot x^* = m$.

With budget slack, a sufficiently small neighbourhood of x^* is still affordable.

LNS provides a strictly better feasible bundle in that neighbourhood: a contradiction.

Therefore an existing optimum spends the budget. → Proof

Convex preferences: what happens to the chosen set?

Take $x, y \in D(p, m)$. The budget is convex, so $z_t = tx + (1 - t)y$ is affordable.
Both x and y are at least as good as every other affordable bundle.
Convex preferences make z_t at least as good as every such bundle too.
Thus $D(p, m)$ is a convex set. This still allows an entire line segment of solutions.

Strict convexity: at most one optimum

Suppose distinct x, y are both optimal.

Their average is affordable, and strict convexity makes it strictly better than either.

That contradicts optimality. Thus, with existence, demand is single-valued.

→ Proof Does strict convexity by itself prove existence?

Indirect utility: the optimal value

$$v(p, m) = \max_{x \geq 0,\, p \cdot x \leq m} u(x).$$

Demand tells us **which bundle**; indirect utility tells us **the attained value**. Even if $D(p, m)$ contains several bundles, every one gives the same value $v(p, m)$. It is a function of the environment, not of a chosen bundle.

How the optimal value changes

Homogeneity: $v(tp, tm) = v(p, m)$.

Prices: raising prices componentwise cannot increase v.

Wealth: under LNS, raising m strictly increases v.

Continuity: continuous u gives continuous v on the maintained positive-price, positive-wealth domain.

A less obvious property: quasiconvexity

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For $0 \leq t \leq 1$, set $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$.

$$v(p^t, m^t) \leq \max\{v(p, m), v(q, n)\}.$$

If a bundle is unaffordable at both endpoints, it is unaffordable at their mixture too.

So the mixed budget cannot deliver utility above both endpoint values. → Proof

Before differentiating, inspect the economics

Substitutes: compare utility gained per unit of spending; a corner may be optimal.

Complements: look for useful proportions; the optimum may be a kink.

Smooth interior problem: marginal conditions can identify a candidate.

Always check feasibility, boundary cases and a reason why the candidate is globally optimal.

Why compare marginal utility per unit of money?

Shift a small amount of spending ε from good j to good i.

$$\Delta x_i = \varepsilon / p_i, \qquad \Delta x_j = -\varepsilon / p_j.$$

The first-order utility change is $\varepsilon\left(u_i/p_i - u_j/p_j\right)$.
If both goods are consumed in a smooth optimum, these marginal returns must agree.

The Lagrangian: one multiplier per inequality

Write the constraints as $m - p \cdot x \geq 0$ and $x_i \geq 0$.

$$\mathcal{L}(x, \lambda, \mu) = u(x) + \lambda(m - p \cdot x) + \sum_i \mu_i x_i.$$

The multipliers satisfy $\lambda \geq 0$, $\mu_i \geq 0$.

For the smooth argument, assume u is differentiable on an open neighbourhood of the candidate.

Stationarity in quantities

At a smooth candidate,

$$u_i(x^*) - \lambda^* p_i + \mu_i^* = 0 \qquad \text{for each } i.$$

One common budget multiplier, but a separate multiplier for every nonnegativity bound.

Stationarity is only one group of conditions. We must also check signs, feasibility and slackness.

The four Kuhn–Tucker groups

Stationarity: $u_i(x^*) - \lambda^* p_i + \mu_i^* = 0.$
Primal feasibility: $p \cdot x^* \leq m$, $x^* \geq 0.$
Dual feasibility: $\lambda^* \geq 0$, $\mu^* \geq 0.$
Complementary slackness:

$$\lambda^*(m - p \cdot x^*) = 0, \qquad \mu_i^* x_i^* = 0.$$

Complementary slackness: read the products

If the budget is slack, $m - p \cdot x^* > 0$, then $\lambda^* = 0$.
If $x_i^* > 0$, then $\mu_i^* = 0$.
If a multiplier is strictly positive, its constraint must bind.
Does a binding constraint always have a strictly positive multiplier?

For consumed goods: equal marginal returns

If $x_i^* > 0$, slackness gives $\mu_i^* = 0$, hence $u_i(x^*) = \lambda^* p_i$.
For two consumed goods with $u_j > 0$,

$$\mathrm{MRS}_{ij} = \frac{u_i(x^*)}{u_j(x^*)} = \frac{p_i}{p_j}.$$

The subjective marginal exchange rate equals the market exchange rate.
This is an interior condition, not a universal rule for every optimum.

For a good at zero: an inequality

If $x_i^* = 0$, stationarity and $\mu_i^* \geq 0$ imply

$$u_i(x^*) \leq \lambda^* p_i.$$

A consumed good has $u_i/p_i = \lambda$; an unconsumed good has $u_i/p_i \leq \lambda$.
At a corner, the unavailable direction of selling a zero holding matters.
Why would a rule that sets every marginal return equal miss some solutions?

A candidate is not yet a solution

Necessity: a differentiable local optimum satisfies KKT under a constraint qualification.

Here $p \gg 0$, $m > 0$ make the active linear constraints regular, including at corners. → Why?

Sufficiency: if u is differentiable and concave, a feasible KKT point is a global optimum.

Quasiconcavity alone does not make every stationary point optimal. → Sufficiency proof → Counterexample

Why saddle points enter the problem

A saddle maximises the Lagrangian in x and minimises it in the nonnegative multipliers:

$$\mathcal{L}(x, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda, \mu).$$

A global saddle gives a feasible optimum and complementary slackness. Concavity links KKT to this global maximisation. → Details

Checkpoint

What established existence? What could establish uniqueness?
Which KKT condition handles a zero quantity?
When is a KKT solution sufficient for global optimality?
What can go wrong if we check only the interior first-order equations?

Example 1: Cobb–Douglas

$$u(x) = x_1^{\alpha} x_2^{1-\alpha}, \qquad 0 < \alpha < 1.$$

A boundary bundle gives zero utility. A strictly positive affordable bundle gives positive utility.

Therefore every optimum is strictly positive. The nonnegativity multipliers are zero.

Now an interior method is justified; it was not assumed at the start.

Choose a convenient representation

Maximising u is equivalent, at the positive optimum, to maximising

$$\widetilde{u}(x) = \alpha \log x_1 + (1-\alpha) \log x_2.$$

$$\mathcal{L} = \alpha \log x_1 + (1-\alpha) \log x_2 + \lambda(m - p_1 x_1 - p_2 x_2).$$

The log representation is concave on the positive domain.

The bundle does not change, but the numerical utility and multiplier generally do.

Solve the interior conditions

$$\frac{\alpha}{x_1} = \lambda p_1, \qquad \frac{1-\alpha}{x_2} = \lambda p_2.$$

Thus $p_1 x_1 = \alpha/\lambda$ and $p_2 x_2 = (1-\alpha)/\lambda$.
Budget exhaustion gives $m = 1/\lambda$, hence $\lambda = 1/m$.

$$x_1(p, m) = \frac{\alpha m}{p_1}, \qquad x_2(p, m) = \frac{(1-\alpha)m}{p_2}.$$

Interpret the solution

$$\frac{p_1 x_1}{m} = \alpha, \qquad \frac{p_2 x_2}{m} = 1 - \alpha.$$

Expenditure shares are constant in this specification.

Demand is homogeneous of degree zero and exhausts the budget.

If $\alpha = 1/2$, $p = (2, 1)$ and $m = 12$, demand is $(3, 6)$.

Which of these predictions is a modelling restriction rather than an accounting identity?

A price counterfactual

Keep $\alpha = 1/2$ and $m = 12$. Raise p_1 from 2 to 3, holding $p_2 = 1$.

What are the new quantities? The spending shares? The attained raw utility?

Does an unchanged quantity of good 2 mean that its utility contribution is unchanged?

Example 2: perfect substitutes

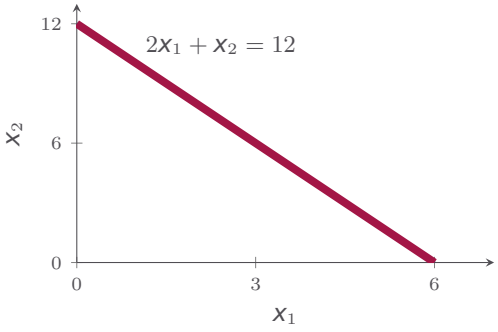
$$u(x) = ax_1 + bx_2, \qquad a, b > 0.$$

Compare a/p_1 and b/p_2: utility gained per unit of spending.
If $a/p_1 > b/p_2$, buy $(m/p_1, 0)$. If the inequality reverses, buy $(0, m/p_2)$.
If the ratios are equal, every point on the budget line is optimal.

Verify a corner with the full KKT system

Let $u(x) = 2x_1 + x_2$, $p = (3, 1)$ and $m = 12$.
The spending returns are $2/3$ and 1, so $x^* = (0, 12)$.
Stationarity and slackness give $\lambda = 1$, $\mu_1 = 1$, $\mu_2 = 0$.
Check: $2 - 3\lambda + \mu_1 = 0$, $1 - \lambda + \mu_2 = 0$.
The zero quantity is optimal precisely because its marginal return is too low.

Equal returns: a whole set of demand



$u = 2x_1 + x_2$, $p = (2, 1)$, $m = 12$.
Every budget-line bundle gives utility 12.
$D(p, m)$ is a segment, while $v(p, m) = 12$ is a single number.

Example 3: perfect complements

$$u(x) = \min\{x_1, x_2\}.$$

To reach utility t, both quantities must be at least t, costing at least $(p_1 + p_2)t$.

$$x_1^* = x_2^* = \frac{m}{p_1 + p_2}.$$

At $p = (2, 1)$ and $m = 12$, the optimum is $(4, 4)$.
The optimum is a kink: the smooth stationarity formula does not apply there.

Example 4: a boundary changes the regime

$$u(a,y) = a + 4\sqrt{y}, \qquad a + y \le m, \quad a, y \ge 0.$$

$$y^* = \min\{m, 4\}, \qquad a^* = m - y^*.$$

If $m > 4$: $a^* > 0$, $\lambda = 1$, and extra wealth goes to a.
If $0 < m < 4$: $a^* = 0$, $\lambda = 2/\sqrt{m} > 1$, $\mu_a = \lambda - 1 > 0$.

A practical solution checklist

1. **Diagnose:** domain, continuity, corners, kinks, satiation.
2. **Establish:** existence, and uniqueness if justified.
3. **Solve:** geometry, a useful substitution or the relevant KKT cases.
4. **Verify:** feasibility, multiplier signs, slackness and global optimality.
5. **Interpret:** quantities, expenditure shares and reactions to parameters.

From a classroom solution to a structural model

Observed $p = (2, 1)$, $m = 12$, $x = (3, 6)$ imply a spending share of $1/2$ on good 1.

Within a Cobb–Douglas specification, this suggests $\alpha = 1/2$.

With the same preference parameter, $p_1 = 3$ predicts $x = (2, 6)$.

Fitting one choice does not establish the functional form. Use more variation and test the restrictions.

Compute indirect utility in the same example

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For the log representation, substitute the demands into $\widetilde{u}$:

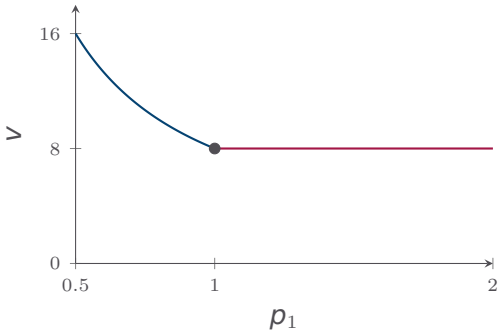
$$v(p, m) = \alpha \log \frac{\alpha m}{p_1} + (1 - \alpha) \log \frac{(1 - \alpha) m}{p_2}.$$

$$v(p, m) = \log m - \alpha \log p_1 - (1 - \alpha) \log p_2 + c_\alpha,$$

where $c_\alpha = \alpha \log \alpha + (1 - \alpha) \log(1 - \alpha)$.

$\rightarrow$ More than two goods

A continuous value need not be differentiable



$u = x_1 + x_2$, $p_2 = 1$, $m = 8$.

$$v(p_1, 1, 8) = \frac{8}{\min\{p_1, 1\}}.$$

At $p_1 = 1$, the consumer is indifferent along the budget line. The value has a kink.

The envelope idea

When the environment changes, the chosen bundle changes too.

At a regular optimum, first-order adjustments of the choice do not add a separate first-order gain to the optimal value.

The envelope theorem reads the value effect from the parameter’s direct entry in the Lagrangian.

We need appropriate regularity and a differentiable value; a kink is a warning.

Differentiate the value, not the choice

At a regular optimum where the envelope formulas apply,

$$v_m(p, m) = \mathcal{L}_m = \lambda^*.$$

For a price,

$$v_{p_i}(p, m) = \mathcal{L}_{p_i} = -\lambda^* x_i^*.$$

The wealth effect is positive when $\lambda^* > 0$; a higher price hurts in proportion to the quantity purchased.

The budget multiplier has an interpretation

$$\lambda^* = v_m(p, m).$$

It is marginal utility from one extra unit of wealth: the value of relaxing the budget.

Its numerical size depends on the chosen utility scale.

A binding budget does not by itself imply $v_m > 0$. → A counterexample

Roy's identity: recover demand from value

Under the maintained consumer assumptions, suppose v is differentiable at (p, m) and $v_m(p, m) > 0$.

$$x_i(p, m) = -\frac{v_{p_i}(p, m)}{v_m(p, m)}.$$

Use $v_{p_i} = -\lambda x_i$ and $v_m = \lambda$; the multiplier cancels.
Under these conditions the optimal quantity vector is unique. → Derivation

Check Roy in the Cobb–Douglas example

$$v = \log m - \alpha \log p_1 - (1-\alpha)\log p_2 + c_\alpha.$$

$$v_m = 1/m, \qquad v_{p_1} = -\alpha/p_1, \qquad v_{p_2} = -(1-\alpha)/p_2.$$

$$x_1 = -\frac{-\alpha/p_1}{1/m} = \frac{\alpha m}{p_1}.$$

Likewise $x_2 = (1-\alpha)m/p_2$. The demand agrees with the direct solution.

Demand survives a change of utility scale

If $\widehat{v} = f(v)$ with $f'(v) > 0$, then

$$\widehat{v}_{p_i} = f'(v) v_{p_i}, \qquad \widehat{v}_m = f'(v) v_m.$$

The common factor cancels from Roy's ratio.

The multiplier changes, but demand does not.

A strictly increasing transformation can have a zero derivative at a point; then the differentiable ratio may be unavailable.

Checkpoint

Which assumptions guarantee existence? Which give uniqueness?
Why can $D(p, m)$ be multivalued while $v(p, m)$ is single-valued?
What replaces equality of marginal returns for an unconsumed good?
What justifies treating a KKT candidate as globally optimal?
What two differentiability/positivity conditions are needed to use Roy's formula?

Read next, and keep the core route clear

Varian: *Intermediate Microeconomics: A Modern Approach*, 9th edition (2014), Chapter 5 *Choice* and Chapter 6 *Demand*.

Advanced: Mas-Colell, Whinston and Green (1995), Chapter 3, section 3.D *The Utility Maximization Problem*; the Mathematical Appendix on constrained optimisation.

The next course block develops expenditure minimisation, Hicksian demand and duality. Reserve slides follow.

Homework 1 is issued after lecture block 3, according to the updated course schedule.

Next block: two ways to ask the consumer question

Utility maximisation: given p, m, how much utility can the consumer attain?
Expenditure minimisation: given $p, \bar{u}$, how little spending attains the target?
One fixes resources; the other fixes a level of the same preferences.

Expenditure minimisation

$$e(p, \bar{u}) = \min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

$$H(p, \bar{u}) = \arg\min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

e is the expenditure function; H is the Hicksian demand correspondence. Assume the target is attainable. A singleton is denoted $h(p, \bar{u})$.

Why an unbounded feasible set can still have a minimum

Assume $p \gg 0$, continuous u, and pick a bundle $\widehat{x}$ with $u(\widehat{x}) \geq \bar{u}$.

Only consider target-feasible bundles with $p \cdot x \leq p \cdot \widehat{x}$.

This set is nonempty, closed and bounded, hence compact.

A continuous cost attains a minimum there; excluded bundles are more expensive. → Details

Hicksian demand: basic properties

Price homogeneity: $H(tp, \bar{u}) = H(p, \bar{u})$ for $t > 0$.

No excess utility: if $\bar{u} \geq u(0)$, every minimiser attains exactly $\bar{u}$.

Convex preferences give a convex minimiser set; strict convexity, with the maintained conditions, gives uniqueness.

No excess utility uses continuity and positive prices, not LNS.

The expenditure function has a different curvature

Homogeneity of degree one: $e(tp, \bar{u}) = te(p, \bar{u})$.

It is nondecreasing in prices and, above $u(0)$ on attainable targets, strictly increasing in the target.

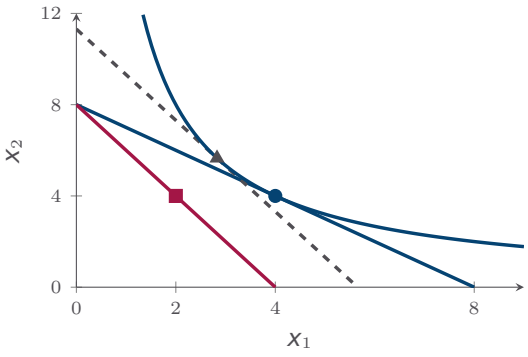
With continuous utility and LNS, expenditure is continuous on the positive-price, attainable-target domain.

Concavity in prices:

$$e(tp + (1-t)q, \bar{u}) \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

It is the infimum of linear cost functions. → Proofs

Fixed wealth versus fixed utility



$u = \sqrt{x_1 x_2}$; initially $p = (1, 1)$, $m = 8$, $x = (4, 4)$.
Raise p_1 to 2: Marshallian demand at the same wealth is $(2, 4)$.
To keep utility 4, spend $8\sqrt{2}$ and buy $(2\sqrt{2}, 4\sqrt{2})$.

Two identities to develop next time

Under continuous LNS preferences, positive prices and an attainable target $\bar{u} > u(0)$:

$$e\big(p, v(p,m)\big) = m, \qquad v\big(p, e(p,\bar{u})\big) = \bar{u}.$$

For unique demands,

$$h\big(p, v(p,m)\big) = x(p,m).$$

These connect the two problems; they are not an assumption that the consumer literally solves both.

A monetary compensation question

Let $u^0 = v(p^0, m)$. How much extra wealth restores this utility after a price change to p^1?

$$\text{CV} = e(p^1, u^0) - m.$$

In the diagram, $\text{CV} = 8\sqrt{2} - 8 \approx 3.31$.

A money-metric change is useful, but summing it across people still involves welfare judgements.

Proof: LNS and budget exhaustion

Suppose an optimum x^* has slack $\delta = m - p \cdot x^* > 0$.
If $\|y - x^*\| < \delta / \|p\|$, then

$$p \cdot y \leq p \cdot x^* + \|p\| \|y - x^*\| < m.$$

LNS supplies a strictly better such y in the consumption domain, contradicting optimality.

→ Back to budget exhaustion

Proof: uniqueness from strict convexity

NES

Suppose distinct $x, y \in D(p, m)$. Their midpoint z is feasible because the budget is convex.

Optimality gives $x \sim y$. Strict convexity of preferences implies $z \succ x$.

This contradicts x being optimal. There is at most one optimum; existence gives a unique one.

→ Back to uniqueness

Proof: indirect utility is quasiconvex

Write $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$ with $0 < t < 1$.
If x violates both budgets, then $p \cdot x > m$ and $q \cdot x > n$, hence $p^t \cdot x > m^t$.
Thus every bundle affordable at the mixture is affordable at at least one endpoint.
Its utility cannot exceed $\max\{v(p, m), v(q, n)\}$. Maximise over the mixed budget.

→ Back to the property

Why the constraint qualification holds here

Active nonnegativity constraints have gradients e_i for the coordinates at zero.

If the budget is slack, these vectors are linearly independent.

If the budget binds, $m > 0$ implies some $x_j > 0$. The budget gradient $-p$ has a nonzero j coordinate, unlike every active bound gradient.

The active gradients are therefore linearly independent. This includes corner optima.

→ Back to necessity and sufficiency

Proof: why concavity makes KKT sufficient

For feasible y, concavity and differentiability imply

$$u(y) - u(x^*) \leq \nabla u(x^*) \cdot (y - x^*).$$

Stationarity gives $\nabla u(x^*) = \lambda^* p - \mu^*$.
Budget feasibility and slackness give $\lambda^* p \cdot (y - x^*) \leq 0$. Bound feasibility and slackness give $-\mu^* \cdot (y - x^*) \leq 0$.
Thus $u(y) \leq u(x^*)$ for every feasible y.

→ Back

Proof: a global saddle is an optimum

Minimisation in multipliers forces feasibility: a violated inequality would let its multiplier send $\mathcal{L}$ to $-\infty$.

It also forces complementary slackness, hence $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$.

For every feasible y,

$$u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*).$$

Thus x^* is optimal.

→ Back to saddle points

A KKT point that is not optimal

NES

Let $u(x) = (x-1)^3$, $p = 1$, $m = 2$, $x \geq 0$.

At $x = 1$, take $\lambda = \mu = 0$. Stationarity, feasibility and complementary slackness all hold.

But $u(2) = 1 > 0 = u(1)$. The KKT point is not optimal.

The function is strictly increasing and quasiconcave. That alone does not make stationarity sufficient.

→ Back

A binding budget with zero marginal value

NES

Use $u(x) = (x-1)^3$, but now $p = 1$, $m = 1$.
The unique optimum is $x^* = 1$: utility is strictly increasing. The budget binds.
Yet $u'(1) = 0$, $\lambda = 0$, and $v_m(1,1) = 0$.
Roy's ratio becomes $0/0$ at that point, although demand $x = m/p$ is perfectly well-defined.

→ Back to the multiplier

Cobb–Douglas with any positive exponent sum

Let $u(x) = \prod_{i=1}^{L} x_i^{a_i}$ with $a_i > 0$ and $A = \sum_i a_i$.
After proving interiority, maximise $\sum_i a_i \log x_i$. Stationarity gives $x_i = a_i/(\lambda p_i)$.
Budget exhaustion gives $\lambda = A/m$, hence

$$x_i(p, m) = \frac{a_i}{A} \frac{m}{p_i}.$$

No restriction $A \leq 1$ is needed for this preference solution.

→ Back

Roy without differentiating demand

Let $x^* \in D(p, m)$. LNS gives $p \cdot x^* = m$.
At any nearby price vector q, the bundle x^* is affordable with wealth $q \cdot x^*$. Thus $F(q) = v(q, q \cdot x^*) \geq u(x^*) = F(p)$.
If v is differentiable, F has a minimum at p:

$$0 = F_{q_i}(p) = v_{p_i}(p, m) + v_m(p, m) x_i^*.$$

If $v_m > 0$, every optimiser satisfies the same Roy formula, so demand is unique.

→ Back

Proof: expenditure minimisation has a solution

Choose an attainable target and a feasible $\widehat{x}$. Let

$$S = \{x \geq 0 : u(x) \geq \bar{u}\} \cap \{x : p \cdot x \leq p \cdot \widehat{x}\}.$$

S is nonempty and compact when u is continuous and $p \gg 0$.
Cost is continuous, so it has a minimum on S. Every omitted feasible bundle costs more than $\widehat{x}$, so this is a global minimum.

→ Back to the preview

Proof: homogeneity and price concavity of expenditure

Scaling prices multiplies every feasible cost by the same positive number:

$$e(tp, \bar{u}) = te(p, \bar{u}), \qquad H(tp, \bar{u}) = H(p, \bar{u}).$$

Let x^t minimise cost at $p^t = tp + (1-t)q$. It is feasible for the target at either price vector.

$$e(p^t, \bar{u}) = tp \cdot x^t + (1-t)q \cdot x^t \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

→ Back to expenditure properties